

Moment generating functions (mgf)

Reference: 1. Meyer PL. Introductory probability and statistical applications. Oxford and IBH Publishing; 1965.
2. Johnson, Richard A., Irwin Miller, and John E. Freund. "Probability and statistics for engineers." (2000).

Definition:

The *moment generating function* of a r.v. X is defined by

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum_i e^{tx_i} p_X(x_i) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & \text{(continuous case)} \end{cases} \quad (4.40)$$

where t is a real variable. Note that $M_X(t)$ may not exist for all r.v.'s X . In general, $M_X(t)$ will exist only for those values of t for which the sum or integral of Eq. (4.40) converges absolutely. Suppose that $M_X(t)$ exists. If we express e^{tX} formally and take expectation, then

$$\begin{aligned} M_X(t) = E(e^{tX}) &= E\left[1 + tX + \frac{1}{2!} (tX)^2 + \cdots + \frac{1}{k!} (tX)^k + \cdots\right] \\ &= 1 + tE(X) + \frac{t^2}{2!} E(X^2) + \cdots + \frac{t^k}{k!} E(X^k) + \cdots \end{aligned} \quad (4.41)$$

and the k th moment of X is given by

$$m_k = E(X^k) = M_X^{(k)}(0) \quad k = 1, 2, \dots \quad (4.42)$$

where

$$M_X^{(k)}(0) = \left. \frac{d^k}{dt^k} M_X(t) \right|_{t=0} \quad (4.43)$$

NOTE:

1. $E(X)$ is coefficient of t in $M_X(t)$.
2. $E(X^2)$ is coefficient of $\frac{t^2}{2!}$ in $M_X(t)$.
3. $E(X^n)$ is coefficient of $\frac{t^n}{n!}$ in $M_X(t)$.
4. $E(X^n) = M_X^n(0)$

Example:

1. Suppose that X has pdf $f(x) = \frac{e^{-|x|}}{2}$, $-\infty < x < \infty$ then find $E(X)$ and $V(X)$ using mgf.

Solution:

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_{-\infty}^{\infty} e^{tx} \frac{e^{-|x|}}{2} dx \\ &= \int_{-\infty}^0 e^{tx} \frac{e^x}{2} dx + \int_0^{\infty} e^{tx} \frac{e^{-x}}{2} dx = \int_{-\infty}^0 \frac{e^{(1+t)x}}{2} dx + \int_0^{\infty} \frac{e^{-(1-t)x}}{2} dx \end{aligned}$$

$$M_X(t) = \frac{1}{1-t^2}, \quad -1 < t < 1$$

$$M_X^1(0) = 0$$

$$M_X^2(0) = 2$$

$E(X) = 0$ and $V(X) = 2$.

$$\text{Or: } M_X(t) = \frac{1}{1-t^2} = 1 + t^2 + t^4 + t^6 + \cdots = 1 + \frac{2t^2}{2} + t^4 + t^6 + \cdots$$

Coefficient of " t " $E(X) = 0$. Coefficient of " $\frac{t^2}{2}$ " $V(X) = 2 - 0 = 2$

2. Let X be a random variable taking the values $0, 1, 2, \dots$ and $f(x) = ab^x$, $a, b > 0$ & $a+b=1$. Find mgf of X . If $E(X) = m_1$, $E(X^2) = m_2$ then S.T $m_2 = m_1(2m_1 + 1)$.

Solution:

$$M_X(t) = \sum_{x=0}^{\infty} ab^x e^{tx} = a \sum_{x=0}^{\infty} (be^t)^x = a \frac{1}{1-be^t} = \frac{a}{1-be^t}$$

$$E(X) = M_X'(0) = \frac{ab}{(1-b)^2}$$

$$E(X^2) = M_X''(0) = \frac{(1+b)ab}{(1-b)^3}$$

Given,

$$E(X) = m_1, E(X^2) = m_2$$

To Prove, $m_2 = m_1(2m_1 + 1)$.

$$\begin{aligned} \text{Consider, } m_1(2m_1 + 1) &= \frac{ab}{(1-b)^2} \left(2 \frac{ab}{(1-b)^2} + 1 \right) = \frac{ab}{(1-b)^2} \left(\frac{2ab+1+b^2-2b}{(1-b)^2} \right) \\ &= \frac{ab}{(1-b)^4} (2b(a-1) + 1 + b^2) = \frac{ab}{(1-b)^4} (2b(-b) + 1 + b^2) = m_2. \end{aligned}$$

Moment generating function for binomial distribution

Let X have the binomial distribution with probability distribution

$$b(x | n, p) = \binom{n}{x} p^x (1-p)^{n-x} \quad \text{for } x = 0, 1, \dots, n$$

Show that

$$(a) \quad M(t) = (1 - p + pe^t)^n \text{ for all } t$$

$$(b) \quad E(X) = np \text{ and } \text{Var}(X) = np(1-p)$$

Solution (a) By definition of the moment generating function

$$\begin{aligned} M(t) &= \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (e^t p)^x (1-p)^{n-x} \\ &= (pe^t + 1 - p)^n \quad \text{for all } t \end{aligned}$$

where we have used the binomial formula

$$(a+b)^n = \sum_{x=0}^n \binom{n}{x} a^x b^{n-x}$$

(b) Differentiating $M(t)$, we find

$$M'(t) = np e^t (pe^t + 1 - p)^{n-1}$$

$$M''(t) = (n-1)np^2 e^{2t} (pe^t + 1 - p)^{n-2} + np e^t (pe^t + 1 - p)^{n-1}$$

Evaluating these derivatives at $t = 0$, we obtain the moments

$$E(X) = np$$

$$E(X^2) = (n-1)np^2 + np$$

Also, the variance is

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = np(1-p)$$



Moment generating function for Poisson distribution

Let X have the Poisson distribution with probability distribution

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda} \quad \text{for } x = 0, 1, \dots, \infty$$

Show that

(a) $M(t) = e^{\lambda(e^t - 1)}$ for all t

(b) $E(X) = \lambda$ and $\text{Var}(X) = \lambda$

The mean and variance of the Poisson distribution are equal.

Solution (a) By definition of the moment generating function

$$\begin{aligned} M(t) &= \sum_{x=0}^{\infty} e^{tx} \frac{\lambda^x}{x!} e^{-\lambda} = \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} e^{-\lambda} \\ &= e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)} \quad \text{for } -\infty < t < \infty \end{aligned}$$

where we have used the series $e^y = \sum_{k=0}^{\infty} \frac{y^k}{k!}$

(b) Differentiating $M(t)$, we find

$$\begin{aligned} M'(t) &= \lambda e^t e^{\lambda(e^t - 1)} \\ M''(t) &= \lambda e^t e^{\lambda(e^t - 1)} + \lambda^2 e^{2t} e^{\lambda(e^t - 1)} \end{aligned}$$

Evaluating these derivatives at $t = 0$, we obtain the moments

$$\begin{aligned} E(X) &= \lambda \\ E(X^2) &= \lambda + \lambda^2 \end{aligned}$$

Also, the variance is

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \lambda$$



Moment generating function for Gamma Distribution

$$f(x) = \begin{cases} \frac{x^{r-1} e^{-\alpha x} \alpha^r}{\Gamma(r)}, & x > 0, \alpha, r > 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \frac{\alpha^r}{\Gamma(r)} \int_0^{\infty} e^{-(\alpha-t)x} x^{r-1} dx$$

Substitute, $x(\alpha - t) = v$ then $dx = \frac{dv}{\alpha - t}$

$$\begin{aligned} M_X(t) &= \frac{\alpha^r}{\Gamma(r)} \int_0^{\infty} e^{-v} \left(\frac{v}{\alpha-t}\right)^{r-1} \frac{dv}{\alpha-t} \\ &= \frac{\alpha^r}{\Gamma(r)(\alpha-t)^r} \int_0^{\infty} e^{-v} (v)^{r-1} dv \\ &= \frac{\alpha^r}{\Gamma(r)(\alpha-t)^r} \Gamma(r) = \left(\frac{\alpha}{\alpha-t}\right)^r \end{aligned}$$

$$\begin{aligned} E(X) &= \frac{r}{\alpha} \\ E(X^2) &= \frac{r(r+1)}{\alpha^2} \\ \text{V}(X) &= \frac{r}{\alpha^2} \end{aligned}$$

Moment generating function for Exponential Distribution

Note: When we sub $r=1$ in gamma distribution we get exponential distribution.

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$M_X(t) = \int_0^{\infty} e^{tx} f(x) dx = \lambda \int_0^{\infty} e^{-(\lambda-t)x} dx$$

$$M_X(t) = \left(\frac{\lambda}{\lambda-t} \right), t < \lambda$$

$$E(X) = \frac{1}{\lambda}$$

$$V(X) = \frac{1}{\lambda^2}$$

Or:

EXAMPLE 10.4. Suppose that X has an *exponential distribution* with parameter α . Therefore

$$M_X(t) = \int_0^{\infty} e^{tx} \alpha e^{-\alpha x} dx = \alpha \int_0^{\infty} e^{x(t-\alpha)} dx.$$

(This integral converges only if $t < \alpha$. Hence the mgf exists only for those values of t . Assuming that this condition is satisfied, we shall proceed.) Thus

$$\begin{aligned} M_X(t) &= \frac{\alpha}{t - \alpha} e^{x(t-\alpha)} \Big|_0^{\infty} \\ &= \frac{\alpha}{\alpha - t}, \quad t < \alpha. \end{aligned} \quad (10.7)$$

Moment generating function for chi square Distribution

Special case of Gamma distribution: $r = \frac{n}{2}$ and $\alpha = \frac{1}{2}$ in Γ function we get χ^2 distribution.

A continuous random variable X is said to have a chi-square distribution if its PDF is given by.

$$f(x) = \begin{cases} \frac{x^{\frac{n}{2}-1} e^{-\frac{x}{2}}}{\Gamma(n/2) 2^{\frac{n}{2}}}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \frac{\alpha^r}{\Gamma(\frac{n}{2}) 2^{\frac{n}{2}}} \int_{-\infty}^{\infty} e^{(t-\frac{1}{2})x} x^{\frac{n}{2}-1} dx \\ M_X(t) &= (1 - 2t)^{-\frac{n}{2}} \end{aligned}$$

$$E(X) = n \text{ and } V(X) = 2n$$

Moment generating function for Uniform Distribution

$$f(x) = \begin{cases} \frac{1}{(b-a)} & a \leq x \leq b \\ 0 & \text{else where} \end{cases}$$

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \frac{1}{b-a} \int_{-\infty}^{\infty} e^{tx} dx \\ &= \frac{1}{t(b-a)} (e^{bt} - e^{at}) \end{aligned}$$

Expand and get the suitable coefficient in the expansion to find $E(X)$ and $V(X)$.

$$\text{Mean } E(X) = \frac{(a+b)}{2}$$

$$E(X^2) = \frac{(a^2 + b^2 + ab)}{3}$$

$$\text{Variance } V(X) = E(X^2) - [E(X)]^2 = \frac{(b-a)^2}{12}$$

Moment generating function for normal distribution

Show that the normal distribution, whose probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} \quad \text{has} \quad M(t) = e^{t\mu + \frac{1}{2}t^2\sigma^2}$$

which exists for all t . Also, verify the first two moments.

By substituting, $v = \left[\frac{x-(t\sigma^2+\mu)}{\sigma^2} \right]$ and $dx = \sigma^2 dv$ and $\int_{-\infty}^{\infty} e^{-x^2} dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Solution:

$$M_X(t) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{tx} \exp\left(-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2\right) dx.$$

Let $(x - \mu)/\sigma = s$; thus $x = \sigma s + \mu$ and $dx = \sigma ds$. Therefore

$$\begin{aligned} M_X(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp[t(\sigma s + \mu)] e^{-s^2/2} ds \\ &= e^{t\mu} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2}[s^2 - 2\sigma ts]\right) ds \\ &= e^{t\mu} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-\frac{1}{2}[(s - \sigma t)^2 - \sigma^2 t^2]\right\} ds \\ &= e^{t\mu + \sigma^2 t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2}[s - \sigma t]^2\right) ds. \end{aligned}$$

Let $s - \sigma t = v$; then $ds = dv$ and we obtain

$$\begin{aligned} M_X(t) &= e^{t\mu + \sigma^2 t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-v^2/2} dv \\ &= e^{(t\mu + \sigma^2 t^2/2)}. \end{aligned}$$

To obtain the moments of the normal, we differentiate once to obtain

$$M'(t) = e^{t\mu + \frac{1}{2}t^2\sigma^2} (\mu + t\sigma^2)$$

and a second time to get

$$M''(t) = e^{t\mu + \frac{1}{2}t^2\sigma^2} [(\mu + t\sigma^2)^2 + \sigma^2].$$

Setting $t = 0$,

$$E[X] = M'(0) = \mu \quad \text{and} \quad E(X^2) = M''(0) = \sigma^2 + \mu^2$$

so $\text{Var}(X) = \sigma^2$ as the notation suggests. ■

A basic property relates the moment generating function of $a + bX$ to that of X .

MGF of some standard distributions:

1. Binomial Distributions: $M_X(t) = M_X(t) = (pe^t + q)^n$
2. Poisson Distributions: $M_X(t) = e^{\alpha(e^t - 1)}$
3. Normal Distributions: $M_X(t) = e^{t\mu + \frac{\sigma^2 t^2}{2}}$
4. Exponential Distributions: $M_X(t) = \frac{\alpha}{\alpha - t}$
5. Gamma Distributions: $M_X(t) = \frac{\alpha^r}{(\alpha - t)^r}$
6. Chi square Distributions: $M_X(t) = (1 - 2t)^{-n/2}$

Properties of mgf

Theorem 10.2. Suppose that the random variable X has mgf M_X . Let $Y = \alpha X + \beta$. Then M_Y , the mgf of the random variable Y , is given by

$$M_Y(t) = e^{\beta t} M_X(\alpha t). \quad (10.12)$$

In words: To find the mgf of $Y = \alpha X + \beta$, evaluate the mgf of X at αt (instead of t) and multiply by $e^{\beta t}$.

Proof

$$\begin{aligned} M_Y(t) &= E(e^{Yt}) = E[e^{(\alpha X + \beta)t}] \\ &= e^{\beta t} E(e^{\alpha t X}) = e^{\beta t} M_X(\alpha t). \end{aligned}$$

Theorem 10.3. Let X and Y be two random variables with mgf's, $M_X(t)$ and $M_Y(t)$, respectively. If $M_X(t) = M_Y(t)$ for all values of t , then X and Y have the same probability distribution.

Theorem 10.4. Suppose that X and Y are independent random variables. Let $Z = X + Y$. Let $M_X(t)$, $M_Y(t)$, and $M_Z(t)$ be the mgf's of the random variables X , Y , and Z , respectively. Then

$$M_Z(t) = M_X(t)M_Y(t). \quad (10.13)$$

Proof

$$\begin{aligned} M_Z(t) &= E(e^{Zt}) = E[e^{(X+Y)t}] = E(e^{Xt}e^{Yt}) \\ &= E(e^{Xt})E(e^{Yt}) = M_X(t)M_Y(t). \end{aligned}$$

Note: This theorem may be generalized as follows: If $X_1, \dots, X_n$ are independent random variables with mgf's M_{X_i} , $i = 1, 2, \dots, n$, then M_Z , the mgf of

$$Z = X_1 + \dots + X_n,$$

is given by

$$M_Z(t) = M_{X_1}(t) \cdots M_{X_n}(t). \quad (10.14)$$

EXAMPLE 10.10. Suppose that X has distribution $N(\mu, \sigma^2)$. Let $Y = \alpha X + \beta$. Then Y is again normally distributed. From Theorem 10.2, the mgf of Y is $M_Y(t) = e^{\beta t} M_X(\alpha t)$. However, from Example 10.8 we have that

$$M_X(t) = e^{\mu t + \sigma^2 t^2 / 2}.$$

Hence

$$\begin{aligned} M_Y(t) &= e^{\beta t} [e^{\alpha \mu t + (\alpha \sigma)^2 t^2 / 2}] \\ &= e^{(\beta + \alpha \mu) t} e^{(\alpha \sigma)^2 t^2 / 2}. \end{aligned}$$

Reproductive Properties of mgf

EXAMPLE 10.11. Suppose that X and Y are independent random variables with distributions $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively. Let $Z = X + Y$. Hence

$$\begin{aligned} M_Z(t) &= M_X(t) M_Y(t) = \exp(\mu_1 t + \sigma_1^2 t^2 / 2) \exp(\mu_2 t + \sigma_2^2 t^2 / 2) \\ &= \exp[(\mu_1 + \mu_2) t + (\sigma_1^2 + \sigma_2^2) t^2 / 2]. \end{aligned}$$

Theorem 10.5 (*the reproductive property of the normal distribution*). Let $X_1, X_2, \dots, X_n$ be n independent random variables with distribution $N(\mu_i, \sigma_i^2)$, $i = 1, 2, \dots, n$. Let $Z = X_1 + \dots + X_n$. Then Z has distribution $N(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2)$.

The Poisson distribution also possesses a reproductive property.

Theorem 10.6. Let $X_1, \dots, X_n$ be independent random variables. Suppose that X_i has a Poisson distribution with parameter α_i , $i = 1, 2, \dots, n$. Let $Z = X_1 + \dots + X_n$. Then Z has a Poisson distribution with parameter

$$\alpha = \alpha_1 + \dots + \alpha_n.$$

Theorem 10.7. Suppose that the distribution of X_i is $\chi_{n_i}^2$, $i = 1, 2, \dots, k$, where the X_i 's are independent random variables. Let $Z = X_1 + \dots + X_k$. Then Z has distribution χ_n^2 , where $n = n_1 + \dots + n_k$.

Proof: From Eq. (10.10) we have $M_{X_i}(t) = (1 - 2t)^{-n_i/2}$, $i = 1, 2, \dots, k$. Hence

$$M_Z(t) = M_{X_1}(t) \cdots M_{X_k}(t) = (1 - 2t)^{-(n_1 + \dots + n_k)/2}.$$

Problems:

1. Find the mgf of random variable X which is uniformly distributed with an interval $(-a, a)$ and hence find $E(X^{2n})$.

Solution: $X \sim U(-a, a) = \frac{e^{at} - e^{-at}}{(a+a)t}$
 $= \frac{e^{at} - e^{-at}}{(2a)t}$, by expanding

we know, $M_X(t) = \frac{1}{t(b-a)} (e^{bt} - e^{at})$

$$E(X^{2n}) = \text{coefficient of } \frac{t^{2n}}{(2n)!} = \frac{a^{2n}}{(2n+1)}$$

2. If X is normally distributed with mean μ and variance, σ^2 then show that $E(X - \mu)^{2n} = 1.3.5 \dots (2n - 1)\sigma^{2n}$.

Solution: Given $X \sim N(\mu, \sigma^2)$, $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$

Let $Y = (X - \mu)$

To get $E(Y^{2n}) =$ The coefficient of $\frac{t^{2n}}{(2n)!}$ in $M_Y(t)$.

$$\begin{aligned} M_Y(t) &= E(e^{ty}) = E(e^{t(x-\mu)}) \\ &= E(e^{tx})E(e^{-\mu t}) \\ &= M_X(t)E(e^{-\mu t}) \end{aligned}$$

$$M_Y(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}} e^{-\mu t} = e^{\frac{\sigma^2 t^2}{2}} = 1 + \left(\frac{\sigma^2 t^2}{2}\right) + \frac{1}{2!} \left(\frac{\sigma^2 t^2}{2}\right)^2 + \frac{1}{3!} \left(\frac{\sigma^2 t^2}{2}\right)^3 + \dots$$

$$E(Y^{2n}) = \text{The coefficient of } \frac{t^{2n}}{(2n)!} \text{ in } M_Y(t) = \frac{\sigma^{2n} (2n)!}{2^n n!} = \frac{(2n)!}{2^n n!} * \sigma^{2n}$$

$$= \frac{(2n)(2n-1)(2n-2)\dots(3)(2)(1)}{(2n)(2n-2)(2n-4)\dots(6)(4)(2)} * \sigma^{2n}$$

$$= 1.3.5 \dots (2n - 1)\sigma^{2n}.$$

3. Let $X_1 \sim \chi^2(3)$, $X_2 \sim \chi^2(5)$ and $Z = X_1 + X_2$ where X_1 and X_2 independent random variables. Find $M_Z(t)$ and $V(Z)$ and pdf of Z .

Solution:

$$X_1 \sim \chi^2(3) \text{ gives } M_{X_1}(t) = (1 - 2t)^{-\frac{3}{2}}$$

$$X_2 \sim \chi^2(5) \text{ gives } M_{X_2}(t) = (1 - 2t)^{-\frac{5}{2}}$$

$$M_Z(t) = M_{X_1}(t)M_{X_2}(t) = (1 - 2t)^{-\frac{8}{2}} \sim \chi^2(8)$$

Therefore, $V(Z) = 2n = 2(8) = 16$. Where $n = 8$.

$$\begin{array}{l} \text{Chi-square} \\ \text{Distribution} \\ X \sim \chi^2(n) \end{array} \left| \begin{array}{l} f(x) = \begin{cases} \frac{x^{\frac{n}{2}-1} e^{-\frac{x}{2}}}{\Gamma(n/2)2^{\frac{n}{2}}}, & x > 0 \\ 0, & \text{elsewhere} \end{cases} \end{array} \right.$$

$$f(Z) = \begin{cases} \frac{Z^{\frac{n}{2}-1} e^{-\frac{Z}{2}}}{2^{\frac{n}{2}} \Gamma(n/2)}, & Z > 0 \\ 0, & \text{Otherwise} \end{cases} = \begin{cases} \frac{Z^3 e^{-\frac{Z}{2}}}{2^4 \Gamma(4)}, & Z > 0 \\ 0, & \text{Otherwise} \end{cases}$$

4. Let X_1, X_2 and X_3 are 3 independent random variable having normal distributions with parameter (4, 1), (5, 2), (7, 3) respectively. Let $Y = 2X_1 + 2X_2 + X_3$. Find the pdf of $V = \left(\frac{Y-\mu}{\sigma}\right)^2$, where μ and σ are the mean and standard deviation of Y.

Solution:

$$M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$Y \sim N(2 * 4 + 2 * 5 + 1 * 7, \quad 2^2 * 1 + 2^2 * 2 + 1 * 3)$$

$$Y \sim N(25, 15)$$

$$M_Y(t) = e^{25t + \frac{15t^2}{2}}$$

We have,

If $X \sim N(\mu, \sigma^2)$, then show that $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ and $Y = Z^2 \sim \chi^2(1)$.

$$\text{Chi-square Distribution } \left. \begin{array}{l} X \sim \chi^2(n) \end{array} \right| f(x) = \begin{cases} \frac{x^{\frac{n}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Therefore, $V = \left(\frac{Y-25}{\sqrt{15}}\right)^2 = Z^2 \sim \chi^2(1)$

$$f(V) = \begin{cases} \frac{e^{-\frac{v}{2}} v^{-\frac{1}{2}}}{2^{\frac{1}{2}} \Gamma(\frac{1}{2})}, & v > 0 \\ 0, & \text{Otherwise} \end{cases} = \begin{cases} \frac{e^{-\frac{v}{2}} v^{-\frac{1}{2}}}{\sqrt{\pi} 2^{\frac{1}{2}}}, & v > 0 \\ 0, & \text{Otherwise} \end{cases}$$

5. If a random variable X has mgf $M_X(t) = \frac{3}{3-t}$ then find the standard deviation of the random variable X.

Solution:

$$M_X(t) = \frac{3}{3-t} = \frac{3}{3(1-\frac{t}{3})} = \frac{1}{(1-\frac{t}{3})} = \left(1 - \frac{t}{3}\right)^{-1} = 1 + \frac{t}{3} + \left(\frac{t}{3}\right)^2 + \dots$$

$$E(X) = \text{coef of } t \text{ in } M_X(t) = \frac{1}{3}$$

$$E(X^2) = \text{coef of } \frac{t^2}{2!} \text{ in } M_X(t) = \frac{1}{9} 2! = \frac{2}{9}$$

$$V(X) = \frac{1}{9}. \text{ Hence, } \sigma = \frac{1}{3}$$

6. Let X be random variable having probability mass function $p(x=k) = p(1-p)^{k-1}$, $k = 1, 2, 3 \dots n$. Find $M_X(t)$ and $V(X)$.

Solution:

$$M_X(t) = \sum_{x=1}^n e^{tx} p(X=x)$$

$$= \sum_{x=1}^n e^{tx} p(1-p)^{x-1}$$

$$= \frac{p}{p-1} \sum_{x=1}^n e^{tx} (1-p)^x$$

$$= \frac{p}{1-p} \sum_1^n (e^t(1-p))^x, \text{ Expanding}$$

$$= \frac{p}{1-p} \{ (e^t(1-p)) + (e^t(1-p))^2 + \dots \}$$

$$= \frac{p}{1-p} (e^t(1-p)) \{ 1 + (e^t(1-p)) + (e^t(1-p))^2 + \dots \}$$

$$M_X(t) = \frac{p}{1-p} * (e^t(1-p)) * \frac{1}{1 - e^t(1-p)} = \frac{pe^t}{1 - e^t(1-p)}$$

$$M_X^1(t) = \frac{pe^t}{[1 - e^t(1-p)]^2},$$

$$\text{at } t=0, E(X) = \frac{1}{p}$$

and

$$E(X^2) = \frac{1}{p} + \frac{2(1-p)}{p^2}$$

$$V(X) = \frac{1-p}{p^2}$$

7. If X has pdf $f(x) = \lambda e^{-\lambda(x-a)}, X \geq a$, find the mgf of X and hence find V(X).

Ans: Given is an exponential distribution.

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_a^{\infty} e^{tx} \lambda e^{-\lambda(x-a)} dx \\ &= \lambda \int_a^{\infty} e^{tx} e^{-\lambda(x-a)} dx \\ &= \lambda e^{\lambda a} \int_a^{\infty} e^{-(\lambda-t)x} dx \\ &= \lambda e^{\lambda a} \frac{e^{-(\lambda-t)x}}{-(\lambda-t)} \\ &= \lambda e^{\lambda a} \frac{e^{-(\lambda-t)a}}{(\lambda-t)} = \frac{\lambda e^{at}}{\lambda-t}, \lambda > t \end{aligned}$$

$$E(X) = a + \frac{1}{\lambda} \text{ and } V(X) = \frac{1}{\lambda^2}$$

8. If the mgf of discrete random variable is $e^{4(e^t-1)}$ then find $P(X = \mu + \sigma)$ where μ and σ are the mean and S.D of X.

Solution:

$$M_X(t) = e^{4(e^t-1)}$$

$X \sim p(\alpha)$ where $\alpha = 4$

Therefore $E(X) = V(X) = 4$,

Which gives $\mu = 4$ and $\sigma = 2$.

$$P(X=k) = \frac{e^{-\alpha} \alpha^k}{k!}, k = 0, 1, 2, \dots, n$$

For, $\alpha = 4$,

$$P(X=k) = \frac{e^{-4} 4^k}{k!}$$

$$\text{To find } P(X = \mu + \sigma) = P(X = 6) = \frac{e^{-4} 4^6}{6!} = 0.1042.$$